

Suhanee Korpe

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Research Interests

A PhD student at Università Bocconi in the department of Computer Science. Interested in Physics, Machine Learning, and Applied Mathematics; interested in pursuing research at the intersection of Statistical Mechanics, Information Theory and Machine Learning / Artificial Intelligence.

Education

Università Bocconi

PhD in Computer Science and Statistics

Milan, Italy

Sept 2026 - Sept 2030

Indian Institute of Science Education and Research, Bhopal

BS-MS Dual Degree

Bhopal, India

Dec 2021 - May 2026

- Score: 9.22/10 [↗](#)

Suryadatta National School

Higher Secondary Education (CBSE)

Pune, India

April 2019 - April 2021

- Score: 95.8% [↗](#)
- Felicitated for achieving the highest marks in Physics in the 12th CBSE Board Exam.

DAV Public School

Primary and Secondary Education (CBSE)

Pune, India

June 2006 - Apr 2019

- Score: 99.2% [↗](#)
- Ranked 1st in Pune district and 2nd in Maharashtra State in 10th CBSE Board Exam.

Projects

Symmetric and Non-Symmetric Matrix Denoising for Non-Rotational Priors

Supervisor: **Dr. Jean Barbier** at the ICTP, Trieste

Trieste, Italy

Bhopal, India

May 2025 – April 2026

- Did theoretical calculations for the free entropy of high-rank symmetric and non-symmetric matrix denoising obtained from the Rademacher and the Gaussian prior by combining replica theory and the HCIZ integrals.
- The goal was to obtain the phase diagram by performing numerical simulations to solve the self-consistent equations and to verify the theory by comparing the theoretical results with the Rotational Invariant Estimator.

Phase Transitions in the Binary Symmetric Perceptron

Supervisor: **Dr. Aurélien Decelle** at the Universidad Complutense de Madrid

Madrid, Spain

May 2024 – Sep 2024

- Studied the symmetric perceptron in two settings. Analytically (via the replica method), computed the free energy of the storage problem and of the teacher-student model.
- Performed numerical calculations in Julia and Python to obtain the phase diagram and characterize phase transitions of the model.

Large Language Models

e-Zest Solutions

Pune, India

Dec 2023

- Studied the transformer architecture and explored techniques for fine-tuning a pretrained LLM (Hugging Face) to translate natural language to graph database queries using PEFT techniques.

Publications

The Symmetric Perceptron: a Teacher-Student Scenario

Jan 2026

G. Catania, A. Decelle, S. Korpe

Physica A: Statistical Mechanics and its Applications, 2026

Conferences & Workshops

- **ICTP-INdAM-SLMATH Summer Graduate School for Machine Learning**

Poster Presentation

ICTP, Trieste

15-27 June 2026

- **Youth in High Dimensions 2025**

Attendee

ICTP, Trieste

July 2025

Skills

Programming Languages: Python (Pandas, NumPy, SciPy, scikit-learn, Matplotlib, Seaborn), Julia, Cypher, SQL, C

Software/Tools: Mathematica, LangChain, LlamaIndex, Neo4j, MATLAB, LaTeX

Languages: English (Proficient), Hindi (Native), Marathi (Native), German (A1 Level)